

Study-Mate – Chat History

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User:

Insertion sort

Assistant:

Algorithm for Insertion Sort:

```
...
Algorithm InsertionSort (a, n)
{
  // sort the array a[1:n] into non decreasing order.
  for i := 2 to n do
  {
    key=a[i];
    // Insert a[i] into the sorted part of the array, i.e., a [1: i -1]
    j=i-1;
    while (j>0 and a[j]>key) do
    {
      a[j+1]=a[j]; // move a[j] to its next position in the right side
      j=j-1;
    }
    a[j+1]=key;
  }
}
...
```

****Performance Analysis:****

Insertion sort takes linear time in the best case and quadratic time in the worst case. The complexity of insertion sort is $O(n^2)$.